DisputeDesk

A dispute lawyer for every small seller, powered by PayPal and AI.

Live demo disputedesk-zcru.onrender.com
Code github.com/kavee05/disputedesk
Judge guide for the PayPal AI Hackathon

A small seller gets a PayPal dispute on a parcel the courier marked as delivered. Winning means an hour of digging through the courier site, PayPal and old emails, then writing a formal reply before a deadline. For a $30 order most sellers simply refund. **DisputeDesk does the digging and the writing, shows its reasoning, and lets the seller decide with one click.** Everything runs on the PayPal Sandbox, so no real money moves.

Try it in two minutes

1. Open the live demo. The free server may take up to a minute to wake if nobody has visited for a while.
2. Open **Pandan leaf placemats** on the Disputes page and press **Run the agent**. Gemini answers in about 15 seconds.
3. Read the four steps. This buyer message tries to give the AI orders, and the agent flags it and argues from the delivery record anyway.
4. Edit the draft if you like, then press **Approve and submit evidence**. The case now says *Processing*, never a guessed result.
5. About 20 seconds later a decision panel appears on its own. Press **Decide for seller** and the case turns to Won.
6. Look at **Impact**, **Policy** and the **Audit log**. **Demo cases** puts the starting data back whenever you want.

No login is needed. The hosted copy is in demo mode with clearly labelled simulated PayPal data and real Google Gemini reasoning. The README explains how to run it against the real PayPal Sandbox with your own keys.

How the agent works

1 Case Analyst

Google Gemini

Reads the numbered evidence from PayPal, estimates the chance of winning, and recommends fight, refund or a human decision.

2 Evidence Writer

Google Gemini

Drafts the reply and cites evidence for every claim. Code then flags any claim that has nothing behind it.

3 Policy Gate

Plain code, no AI

Checks the seller's limits on amount, confidence and dispute reason. It allows the step, asks the seller, or blocks it.

4 Executor

PayPal Disputes API

Reads the dispute again, sends only allowlisted actions PayPal still offers, and stores PayPal's answer exactly as received.

Before deciding, the app weighs the two options with the seller's own numbers. If p is the chance of winning, A the disputed amount and T the cost of the seller's review time, refunding costs **A** while fighting costs **(1 − p) × A + T** on average. The case page draws this as a balance scale.

What is real and what is simulated

Part	Live demo	With PayPal Sandbox keys
Disputes, orders, payments, tracking	A simulator that copies the sandbox behaviour we measured, labelled Simulated on every case	Real PayPal Sandbox REST APIs
Evidence, messages, refunds	Simulated, with the same rules the sandbox enforces	Real API calls, responses stored word for word
Final decision on a case	Simulated PayPal decision, labelled	Sandbox adjudication, a PayPal testing tool, labelled
AI reasoning and drafting	Real Google Gemini	Real Google Gemini
Hours saved	An estimate from the seller's settings, and labelled as one	

What we proved on the real PayPal Sandbox

Before building any screens we ran probe scripts against the real sandbox, opened four disputes as a test buyer, and recorded every response. The full log is in FEASIBILITY.md.

Case	What we did through the API	What PayPal confirmed
$48 tote bag	Asked for seller evidence, submitted tracking as evidence, then used sandbox adjudication for the seller	Won Held funds released and the dispute resolved for the seller about two and a half minutes later
$32 silk scarf	Sent a message, escalated to a claim, submitted evidence	Under review Left open on purpose as an unfinished case
$15 candle set	Accepted the claim	Refunded Refund marked completed about two minutes before the dispute record caught up
$27 placemats	Escalated, submitted evidence, sandbox adjudication for the buyer	Refunded Refund of $27 completed and the payment marked refunded

What the sandbox taught us

Nothing settles instantly

Every action returns success at once, then the dispute sits in review for a few minutes. The app shows *Processing at PayPal* and checks again instead of assuming the result.

Typed evidence was refused

The sandbox rejected tracking sent as proof of fulfilment, even though it had asked for it. Evidence of type Other works, so the tracking details travel in the notes.

The refund is the real record

On the candle case the refund was complete two minutes before the dispute said so. The app now trusts the refund itself, and only that, before it shows Refunded.

The decision tool takes its time

The sandbox adjudication option appeared 18 to 28 minutes after evidence went in. Until then a case honestly reads *Under review at PayPal*.

Rules the code enforces

1. **Status comes from PayPal**
   The AI can recommend and draft. It can never mark a case won or refunded.
2. **Buyer text is never an instruction**
   Messages are wrapped as untrusted claims, and any attempt to steer the agent sends the case to the seller.
3. **Personal data stays home**
   Names, emails, phones and card numbers are removed before the model sees anything. Tracking numbers stay, because they are the evidence.
4. **The seller stays in charge**
   The agent only acts alone on small, clear cases, never on unauthorised payment claims, and every draft can be edited first.
5. **Secrets stay on the server**
   Keys live in environment variables, never enter a prompt, and are stripped from the audit log.
6. **Simulation never poses as PayPal**
   In demo mode every message says Simulated PayPal. A test checks that this can never slip.

Built with

PayPal Disputes API, Orders API, Payments API for captures and refunds, Shipment Tracking API, all on the free sandbox. The Agent Toolkit was tested for reading disputes.

AI Google Gemini 3.5 Flash Lite, with Gemini 3.8 Flash as a backup, returning schema checked JSON. Claude Code helped us build and test the project.

App Next.js, TypeScript, Tailwind CSS, SQLite with Drizzle, zod, and 33 automated tests. Hosted on Render.

> **Want the real PayPal path?** Add your sandbox keys to a local copy as the README shows. The banner turns blue, and every status then comes straight from a PayPal response.